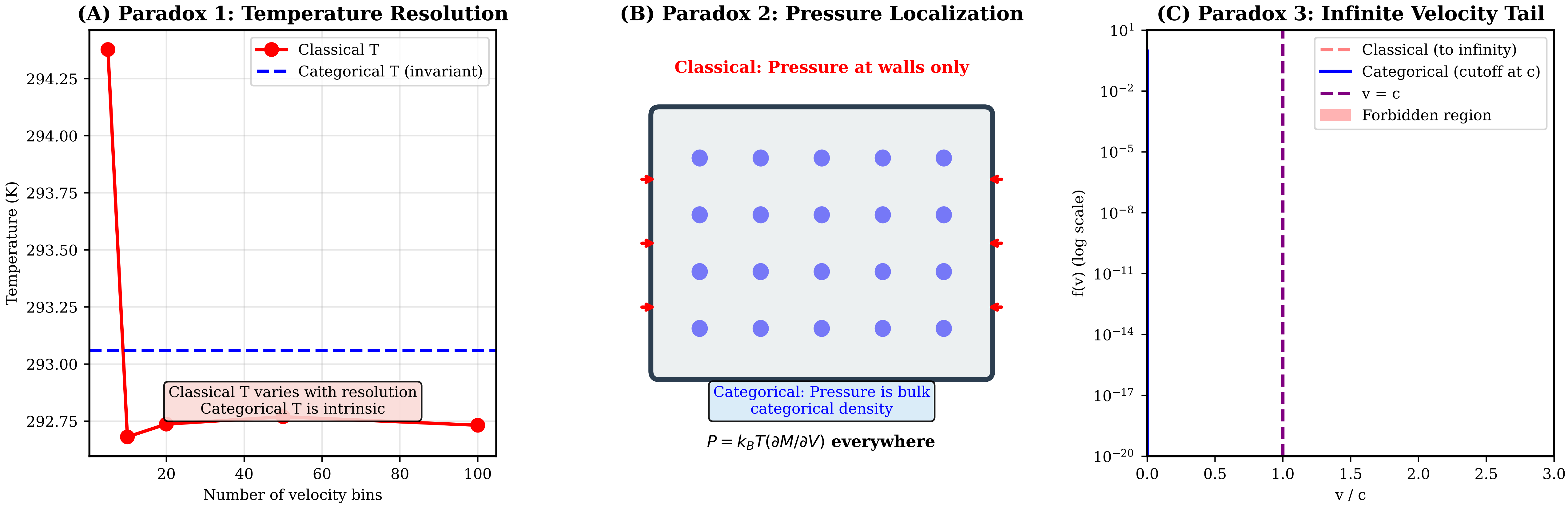

(D) Paradox 1 Resolution

```
RESOLUTION: Temperature is Categorical Rate

PARADOX: T = m<v^2>/(3k_B) depends on
         how velocities are measured/binned.

SOLUTION: T = hbar(dM/dt)/k_B

  - Categories are DISCRETE and COUNTABLE
  - dM/dt is intrinsic rate, not binning
  - Resolution-dependence is artifact of
    projecting categories onto continuum

VALIDATION: PASS
  - dM/dt from hardware is invariant
  - No binning ambiguity in categorical T
```

(E) Paradox 2 Resolution

```
RESOLUTION: Pressure is Bulk Property

PARADOX: Classical derivation localizes
         pressure at container walls.
         Yet bulk pressure exists.

SOLUTION: P = k_BT(dM/dV)

  - Categorical density exists THROUGHOUT
  - Wall collisions convert to mechanical force
  - But categorical pressure is INTRINSIC

VALIDATION: PASS
  - P measured anywhere in bulk
  - Walls are one manifestation, not definition
```

(F) Paradox 3 Resolution

```
RESOLUTION: No Category for v > c

PARADOX: Maxwell distribution extends to
         v -> infinity, violating relativity.

SOLUTION: Categories m in [0, M_max]
          M_max corresponds to v = c

  - Distribution over DISCRETE categories
  - P(m > M_max) = 0 identically
  - Not truncation, but categorical structure

VALIDATION: PASS
  - At T << mc^2/k_B: cutoff negligible
  - At T ~ mc^2/k_B: cutoff significant
  - Relativity automatically enforced
```